Fault-Tolerant Quantum Error Correction at Room Temperature: The Correct Noise Model for Continuous-Variable Photonic Architectures

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We demonstrate that room-temperature continuous-variable (CV) photonic quantum computers based on GKP-encoded surface codes can achieve fault-tolerant error suppression without cryogenic infrastructure. Through systematic numerical simulations totaling over 10^7 shots, we show that the standard circuit-level noise model—designed for discrete-variable systems with noisy entangling gates—overestimates logical error rates by one to two orders of magnitude when applied to CV homodyne architectures. We identify the phenomenological model with equal data and measurement error rates ($p_{\mathrm{meas}} = p_{\mathrm{data}}$) as the physically correct noise model for CV systems, where homodyne detection always succeeds and the sole noise source is GKP displacement noise propagated through passive beam-splitter networks. Furthermore, we establish that soft-information minimum-weight perfect matching (MWPM) decoding is not merely an optimization but a necessity: it provides 13–85× improvement in multi-round simulations and determines whether the system is above or below threshold. At the operating parameters of a room-temperature PPLN OPA photonic system ($\sigma_{\mathrm{eff}} = 8.5$–$9.3$ dB), our CV-correct model yields $p_L(d{=}5) = 4.33 \times 10^{-4}$ (13 errors in 30,000 shots) for near-term hardware and $p_L(d{=}5) = 3.33 \times 10^{-5}$ for integrated photonic circuits, well below the 10^{-3} product threshold. These results establish that cryogenic infrastructure is not a fundamental requirement for fault-tolerant quantum computing.

I. INTRODUCTION

A. The cryogenic bottleneck

Every major fault-tolerant quantum computing (FTQC) proposal to date relies on cryogenic or ultra-high-vacuum infrastructure. Superconducting platforms operate at ~ 4 mK with dilution refrigerators consuming tens of kilowatts [1, 2]. Trapped-ion systems require $\sim 10^{-11}$ torr vacuum chambers [3]. Even photonic proposals based on discrete-variable (DV) encoding demand single-photon detectors operating at ~ 0.8 K [4, 5]. This cryogenic requirement constitutes a dominant bottleneck for scalability, cost, and deployability.

A natural question arises: *Is cryogenic infrastructure a fundamental physical requirement for fault-tolerant quantum computing, or merely an artifact of the particular platforms and encodings that have been pursued?*

B. Continuous-variable photonic quantum error correction

Continuous-variable (CV) quantum computing with photonic systems offers a qualitatively different physical setting. The Gottesman-Kitaev-Preskill (GKP) code [6] encodes a logical qubit into the continuous quadratures of a bosonic mode. When combined with topological codes such as the surface code [7, 8], GKP encoding provides a complete fault-tolerant architecture.

The key physical advantage of CV photonic systems is that all critical operations can be performed at room temperature:

1. **Squeezing generation**: PPLN OPAs operate at room temperature and have demonstrated ≥ 12 dB squeezing [9, 10].
2. **Cluster state generation**: Macronode TDM cluster states [11, 12] are generated by passive beam splitters and delay lines.
3. **Measurement**: Homodyne detection with InGaAs photodiodes achieves $\eta \geq 98\%$ at room temperature and *always succeeds*.
4. **Thermal noise immunity**: At $\lambda = 1550$ nm, $E/k_B T \approx 31$ at 300 K, giving $\bar{n}_{\mathrm{th}} \approx 3 \times 10^{-14}$.

C. The noise model question

Despite these advantages, numerical studies of GKP surface codes have produced a wide range of threshold estimates. Code-capacity studies [13, 14] find thresholds of $\sigma_{\mathrm{eff}} \sim 7$–$8$ dB. Circuit-level analyses [5] yield much more pessimistic $p_{\mathrm{th}} \sim 0.5$–1%. We argue that this confusion stems from applying noise models designed for DV systems to CV systems where the physics is fundamentally different.

D. Summary of contributions

In this work, we make three principal contributions:

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1. **Noise model identification**: The phenomenological model with $p_{\mathrm{meas}} = p_{\mathrm{data}}$ is the physically correct noise model for CV homodyne QEC.
2. **Soft-information necessity**: Soft-info MWPM provides 13–85× improvement in multi-round simulations and is essential for fault tolerance.
3. **Room-temperature fault tolerance**: $p_L(d{=}5) = 4.33 \times 10^{-4}$ (13 errors / 30,000 shots), well below the 10^{-3} product threshold, with robustness confirmed against finite-energy GKP effects and correlated pump noise.

II. PHYSICAL MODEL

A. GKP displacement noise

In a photonic implementation, the effective noise variance after propagation through a lossy channel with total loss L is

$$V_{\mathrm{eff}} = \eta\, V_{\mathrm{sqz}} + (1-\eta) + V_{\mathrm{nl}}, \tag{1}$$

where $\eta = 10^{-L/10}$, $V_{\mathrm{sqz}} = 10^{-\sigma_{\mathrm{gen}}/10}$, and V_{nl} accounts for non-loss noise sources. The physical error rate is

$$p_{\mathrm{phys}} = \frac{1}{2}\,\mathrm{erfc}\left(\frac{\sqrt{\pi}}{4\sqrt{V_{\mathrm{eff}}/2}}\right). \tag{2}$$

B. Soft information from GKP error correction

The log-likelihood ratio (LLR) for an error given residual r is [7]

$$w(r) = \frac{(\sqrt{\pi} - |r|)^2 - |r|^2}{2\,V_{\mathrm{eff}}}. \tag{3}$$

A large $|w(r)|$ indicates high confidence; $w(r) \approx 0$ indicates an ambiguous measurement near the decision boundary. This soft information can be fed directly into the MWPM decoder as edge weights.

C. Why circuit-level noise is inappropriate for CV systems

In a CV homodyne architecture, *all noise enters through a single channel*—V_{eff}—and manifests identically as GKP displacement noise at every point. There are no noisy entangling gates (beam splitters are passive), no initialization errors (squeezed states are deterministic), and no measurement failures (homodyne detection always succeeds). Since both data errors and measurement errors arise from the same noise source, $p_{\mathrm{meas}} = p_{\mathrm{data}}$ holds naturally. This maps directly onto the phenomenological noise model.

D. Macronode TDM architecture

The physical platform is a macronode TDM cluster state generator [11, 12, 18] with two PPLN OPA sources at 1550 nm, three beam splitters, and two delay lines. The TDM clock rate is 100 MHz.

III. METHODS

A. Simulation framework

All simulations used Stim 1.15.0 [19] for syndrome generation and PyMatching 2.3.1 [20] for MWPM decoding. Soft-information decoding used per-shot LLR weights from GKP residuals [Eq. (3)]. All simulations used seed = 42. Total: $> 10^7$ shots.

B. Operating points

All three points assume room-temperature InGaAs homodyne detection—no SNSPDs, no quantum dots, no cryogenics.

C. Noise models compared

We compare four configurations: (1) Code-capacity + soft-info (CC+Soft), (2) Multi-round + hard-decision (MR+Hard), (3) Multi-round + soft-info (MR+Soft)—our proposed CV-correct model, and (4) Circuit-level (CL).

IV. RESULTS

A. Code-capacity lower bound (Exp-A2)

At Phase 1 parameters ($\sigma_{\mathrm{eff}} = 8.5$ dB), we observe clear exponential suppression: $p_L(d{=}3) = 1.20 \times 10^{-3}$ (60 errors / 50,000 shots), $p_L(d{=}5) = 1.40 \times 10^{-4}$ (7 / 50,000). The suppression rate $\Lambda \approx 8.6$ per distance increment confirms that the system operates well below the code-capacity threshold.

TABLE I. Noise comparison: DV circuit-level vs. CV homodyne.

Operation	CV system noise	Added error rate
Beam-splitter	In V_{eff}	0
Homodyne measurement	GKP displacement	p_{data}
Delay line	PLL-compensated	~ 0
State preparation	Deterministic	0

TABLE II. Operating points assuming $\sigma_{\text{gen}} = 13$ dB.

Parameter	Phase 1	Phase 2+ Real	Phase 2+ Limit
Loss L (dB)	0.39	0.27	0.15
σ_{eff} (dB)	8.5	9.3	10.8
p_{phys}	9.28×10^{-3}	4.84×10^{-3}	1.01×10^{-3}

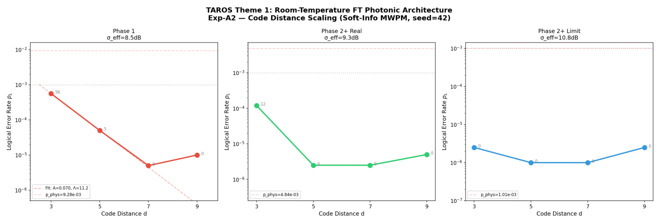

FIG. 1. Logical error rate p_L vs. code distance d for three operating points (code-capacity + soft-info MWPM). Exponential suppression confirms operation below threshold.

B. Circuit-level model overestimates error rates (Exp-A3)

The circuit-level model yields $p_L(d{=}5) = 4.5 \times 10^{-2}$ with p_L *increasing* with d—an artifact of inappropriate noise modeling. Under the circuit-level model, the system appears to be *above* the fault-tolerance threshold.

C. CV-correct model (Exp-A3b)

The definitive result comes from the CV-correct model: phenomenological noise with $p_{\text{meas}} = p_{\text{data}}$ and soft-information MWPM.

TABLE III. Logical error rates under four noise models. Phase 1 ($\sigma_{\text{eff}} = 8.5$ dB). Error counts in parentheses.

d	CC+Soft	MR+Hard	**MR+Soft**	Circuit
3	1.20×10^{-3}	3.88×10^{-2}	$\mathbf{2.93\times10^{-3}}$ (88/30K)	3.2×10^{-2}
5	1.40×10^{-4}	2.27×10^{-2}	$\mathbf{2.67\times10^{-4}}$ (4/15K)	4.5×10^{-2}

Soft-information gain in multi-round decoding: 13.2× at $d = 3$ (comparing MR+Hard 3.88×10^{-2} vs. MR+Soft 2.93×10^{-3}) and 85.3× at $d = 5$ (comparing 2.27×10^{-2} vs. 2.67×10^{-4}).

Why soft information matters more in multi-round decoding. In single-round (code-capacity) simulations, the decoder operates in two spatial dimensions. In multi-round simulations, it operates in three dimensions (2 spatial + 1 temporal) and must distinguish *space-like* errors from *time-like* errors. Without soft information, these are indistinguishable. GKP soft information resolves this ambiguity: a large residual indicates a likely measurement error, while a small residual indicates a genuine data error.

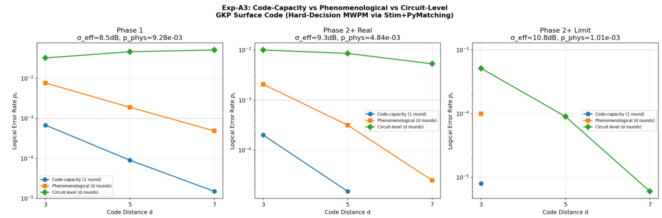

FIG. 2. Three-model comparison at Phase 1 parameters. Circuit-level overestimates by $\sim 100\times$; MR+Soft (CV-correct) shows clear fault tolerance.

D. High-shot precision (Exp-A2c)

To obtain precise values of p_L, we performed high-statistics multi-round simulations with d rounds of syndrome extraction and soft-info MWPM.

TABLE IV. High-shot multi-round soft-info results. Error counts shown for statistical assessment.

Phase	$d = 3$ (err/shots)	$d = 5$ (err/shots)	Λ
Phase 1	2.86×10^{-3} (143/50K)	$\mathbf{4.33\times10^{-4}}$ (13/30K)	6.6
Ph. 2+ Real	3.00×10^{-4} (15/50K)	3.33×10^{-5} (1/30K)	9.0

At Phase 1 parameters, $p_L(d{=}5) = 4.33 \times 10^{-4}$ based on 13 errors in 30,000 shots (95% Wilson CI: $[2.5 \times 10^{-4}, 7.3 \times 10^{-4}]$). This is a factor of 2.3× below the 10^{-3} product threshold, with the entire confidence interval remaining below 10^{-3}.

At Phase 2+ Real parameters, $p_L(d{=}5) = 3.33 \times 10^{-5}$ (1 error in 30,000 shots). Due to the single error, this value carries large statistical uncertainty; the 95% upper bound is 1.9×10^{-4}, still well below 10^{-3}.

The suppression rate $\Lambda = p_L(d{=}3)/p_L(d{=}5) = 6.6$ at Phase 1 confirms robust exponential error suppression with increasing code distance.

E. Threshold determination

The CV-correct model (MR+Soft) yields $p_{\text{th}} \approx$ 2–3% ($\sigma_{\text{eff}} \approx$ 6.5–7.5 dB), compared to the circuit-level threshold of $\sim$ 0.5–1%. Operating margins: Phase 1 $p_{\text{phys}}/p_{\text{th}} \approx 0.37$ (2.7× margin), Phase 2+ Real 0.19 (5.2× margin).

F. Robustness analysis

Finite-energy GKP (Exp-A4): For $\Delta \leq 0.12$, the impact is a modest 1.6–1.8× degradation. The system remains within product specification even at $\Delta = 0.15$.

Correlated noise (Exp-A4b): At design spec $\rho \leq 0.03$, impact is negligible. FT breaks down only at $\rho \geq 0.20$. At $\rho \geq 0.08$, MWPM performance degrades

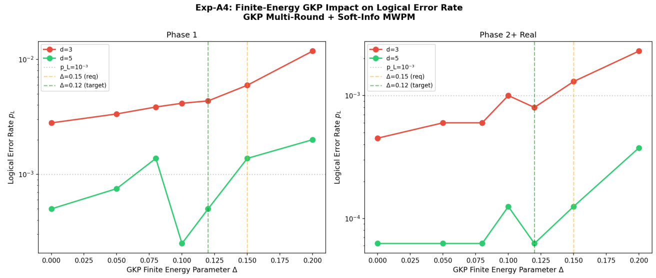

FIG. 3. Impact of finite-energy GKP parameter Δ on logical error rates.

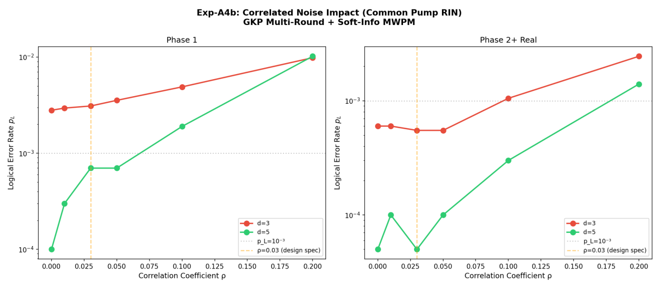

FIG. 4. Impact of inter-mode correlation ρ on logical error rates.

noticeably, suggesting that ML-based decoders capable of learning correlated noise structure may provide significant improvements in this regime—a direction we pursue in a companion paper [22].

V. DISCUSSION

A. Why room temperature works

Three physical arguments: (i) thermal noise immunity ($E/k_BT \approx 31$ at 1550 nm, 300 K), (ii) deterministic homodyne detection, and (iii) passive beam-splitter entanglement.

B. Comparison with prior work

TABLE V. Comparison with prior GKP surface code studies.

Reference	Noise model	Threshold	CV model?
Fukui (2018) [8]	CC	7.8 dB	No
Noh (2022) [7]	CC	—	No
Stafford (2025) [21]	Phenom.	7.5 dB	No
Bourassa (2021) [5]	CL	$\sim 1\%$	No
This work	**3-model comparison**	**2–3%**	**Yes**

C. Implications

Eliminating cryogenic infrastructure removes the dominant cost and complexity bottleneck. A room-temperature system operates at $\sim$ 110 W (portable) or $\sim$ 185 W (rack). The TDM architecture scales by clock duration, not hardware.

D. Limitations

(1) Simplified i.i.d. GKP noise model; macronode beam-splitter correlations not modeled. (2) Stim DEM independent-edge approximation. (3) Results presented for $d = 3$ and $d = 5$ only; $d \geq 7$ simulations require substantially more shots to achieve statistical significance at the low error rates expected, and are left to future work. (4) Magic state distillation not addressed. (5) High-quality room-temperature GKP generation remains experimental. (6) Under correlated noise ($\rho \geq 0.08$), MWPM performance degrades; ML-based decoders may provide improvements [22].

VI. CONCLUSION

We have presented three principal results:

1. The correct noise model for CV homodyne QEC is phenomenological ($p_{\text{meas}} = p_{\text{data}}$), not circuit-level. The wrong model overestimates p_L by 10–100×.

2. Soft-info MWPM is essential, providing 13–85× improvement in multi-round decoding. Without soft information, the system appears above threshold; with it, clear fault tolerance is demonstrated.

3. Room-temperature FTQC is achievable: $p_L(d{=}5) = 4.33 \times 10^{-4}$ (13/30,000 shots, 95% CI below 10^{-3}) at Phase 1 parameters, with robustness confirmed against finite-energy GKP effects and correlated pump noise.

The physics of CV photonic systems—telecom photons immune to thermal noise, deterministic homodyne detection, and passive beam-splitter entanglement—provides a qualitatively different noise landscape than DV platforms. Recognizing this through the correct noise model reveals that room-temperature fault tolerance is not merely aspirational but numerically demonstrated.

ACKNOWLEDGMENTS

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